

PHY 101 PRACTICE QUESTIONS (NOT FOR SALE)

INSTRUCTIONS: Practice below questions religiously, you will find them useful. Try your best to solve them. Consult your colleagues, seniors, text books and internet resources if necessary. Also, revise notes for definitions, discussions/explanations, examples etc. Work hard and make PAAU and your parents proud. Avoid Malpractice of all kinds before, during and after Examinations.

Wishing You the Very BEST in Your Academic Pursuit.

1. A projectile is fired with initial velocity of 100m/s at an angle of 30° with the horizontal. Find the (a) time of flight (b) maximum height attained © range
2. A ball of mass 0.3 Kg, moving at a velocity of 20m/s is suddenly hit by a force of 5 N for a time of 0.03s. find its new velocity of motion.
3. State the three newton laws of motion
4. Three vectors are given by $A = i + 4j - 2k$, $B = -3i + 7j + k$, and $C = 4i + 2j + 3k$, Find $A \cdot (B + C)$
5. Total internal reflection occurs when light moves from amedium to amedium respectively
6. In a liquid of density 13500 kg/m³, longitudinal wave with frequency 200 Hz are Found to have wavelength of 4cm. Find the bulk modulus of the liquid.
7. A 100.0 watt immersion heater is used to heat 200.0 g of water from 10.0 degree C to 70° C. How many minutes did this take? ($c = 1.00 \text{ J/kg/}^{\circ}\text{C}$)
8. The average human body temperature is 98.6 Degree F. What is the equivalent temperature on the Celsius scale?
9. A force of 20N acts on an area of 10m², what is the pressure exerted on the surface?
10. Calculate the pressure exerted on the ground by a person weighing 50kg whose two feet cover an area of 200 cm² ($g = 10 \text{ m/s}^2$)
11. The Cartesian representation of the polar points $(20, \pi/6)$ is:
12. for what values of x and y will $M^x L^y T^z$ be the dimension of surface tension?
13. The distance travelled by a particle in time t seconds is described by the equation $X = 5t + 12t^2$. Find the average speed of the particles between the time interval $t = 2s$ and $t = 6s$.

14. For a medium of refractive index, $n=1.33$, the critical angle of light through it is approximately....degrees
15. An FM station broadcasts at a radio frequency of 90.5Mhz, what is the wavelength of the radio frequency?
16. Three vectors are given by $P = -2i - 5j + 2k$, $Q = -4i - 2j - 3k$ and $R = -3i + 7j + k$, Determine $Q \times R$
17. If the fundamental frequency of a plucked wire is 200KHz, what is the sixth harmonic frequency of the string
18. For what values of x , y and z will $M^x L^y T^z$ be the dimension of momentum?
19. A tank of 2.5m deep is full of water. Calculate the pressure at the bottom of the water tank due to the liquid.
20. The distance X covered by a particle at time t seconds is given as $X = 2t^4 + 10t^3 - 12t^2 - 10t$. find its speed in the 10th second
21. The distance X covered by a particle at time t seconds is given as $Y = t^4 + 5t^3 - 2t^2 - 4t$. find the acceleration at $t = 5s$
22. An aircraft covers a distance of 450km; 60 degree East of south and then flew 550km, 45 degree east of north. Find the resultant displacement.
23. For a closed pipe, the length L of the resonating air column is found to be 15.0cm when fork of frequency $f = 412\text{Hz}$. Find velocity of sound
24. A boy walks 25km north and then 15km in the direction 40 degree East of North. Calculate the distance from his starting point.
25. Three vectors are given by $A = 4i + 3j - 4k$, $B = -i - 4j + 6k$, and $C = 2i + 5j + k$, find $A \cdot (B + C)$
26. A body on a mango tree falls to Y cm from a height of 600cm. How fast is it travelling when it is 4m from the ground? Assume the body has a mass of 30kg.
27. Define: Space, Vector, Scalar, Energy, Work, Pressure, Momentum, Impulse, RADAR
28. A 4kg mass is attached to the end of a wire of length 20m and diameter 0.54mm with an extension of 0.59mm, Find the tensile stress

29. The Cartesian representation of the polar points $(10, 60^\circ)$ is:
30. A car travels at 10m/s , sounds its horn which has a frequency of 500Hz . What frequency is heard by a stationary distant observer as the car approaches?
31. In a liquid of density 500kg/m^3 , longitudinal wave with frequency 10kHz are found to have wavelength of 4m . calculate the bulk modulus of the liquid.
32. A gas does 42J of work while expanding adiabatically, what is the change in internal energy?
33. What is the Lower limit for **Fahrenheit**, **Kelvin** and **Celsius** scales respectively?
34. A force of 700N gives an object of unknown mass acceleration of 10 m/s^2 . Find the mass of the object.
35. A plane progressive wave is represented by the equation $y = 55\text{mm} \sin (7x - 2000t)$ what is the **wavelength, amplitude and speed** of the wave?
36. In the Hooke's Law equation, $F = -Kx$, the minus sign indicates what?
37. A man of mass 75kg walks up a flight of 20 steps each 100mm high in 10 seconds. Find the average power developed by the man ($g = 10\text{m/s}^2$)
38. A golf ball weighs 50g . if its velocity immediately after been driven is 200m/s , what was the impulse of the blow?
39. Find the resultant of a force 10N acting at right angles to another force of 20N .
40. Given that the Young modulus of steel $= 2 \times 10^{11} \text{ N/m}^2$, density of steel $= 10000 \text{ kg/m}^3$, bulk density of water $= 2.1 \times 10^9 \text{ N/m}^2$. What is the Velocity of sound in steel?
41. What do you understand by Conduction, Convection and radiation of heat
42. A constant horizontal force of 4.5N is applied to 2kg block initially at rest on a friction free surface (friction-less). How much work is done on the block in the first 3 seconds after the force is applied.
43. A train is travelling at 65m/s in still air. The frequency of the note emitted by the locomotive whistle is 200Hz (Hint: speed of sound in air $= 340 \text{ m/s}$) Find the wavelength of the sound wave in **front of** and **behind** the locomotive.

44. A train is travelling at 60 m/s in still air. The frequency of the note emitted by the locomotive whistle is 500Hz (Hint: speed of sound in air = 340 m/s) What is the frequency of sound heard by a stationary listener (a) in front of the locomotive (b) behind the locomotive?
45. The distance between the upper and lower fixed point of a thermometer is 40cm. What temperature in Celsius is recorded when the thread is 10 cm above the ice point
46. A crane lifts a load of weight 5000N through a height 20m in 15 seconds. What is the power of the crane in watts and kilowatts?
47. The Cartesian coordinates of a point are given by $(x, y) = (-17, -19)$ m. Find the polar coordinates of this point
48. The position of an object moving in a straight line is given by $x = 3.0t - 4.0t^2 + t^3$, where x is in meters and t is in seconds. What is the position of the object at $t = 2$ s
49. The volume of a fixed mass of gas at constant pressure is 150 cm^3 at 0°C and 245 cm^3 at 150°C . Find the cubic expansivity of the gas.
50. Determine the distance between consecutive antinodes (A) of a string if the wavelength is 20cm.
51. Convert $(15, \pi/5)$ to cartesian coordinate. Note that π is 180°
52. List three fundamental quantities and their units
53. List three derived quantities and their units
54. When the brakes are applied to a moving car travelling at 60 km/hr, it decelerates at a uniform rate of 5 m/s^2 . calculate the time taken to reach a velocity of 36 Km/hr
55. A ball is thrown vertically upwards with an initial velocity of 40m/s. find (a) its velocity after 3 secs (b) maximum height attained and the time taken to reach it (C) total time taken for the ball to return to the ground.